



Analyzing Bat Activity in Burned Red Pine Stands in Northern Wisconsin

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Introduction

- Bat populations in Wisconsin are declining.
 - Threats include white-nose syndrome, wind turbines, and habitat loss.
- Fire has historically disturbed landscapes in Wisconsin.
 - Many ecosystems have evolved with it.
- Routine fires thin out tree and understory cover.
- Open forests influence bat community dynamics, locomotion, and foraging styles.
- **Objective:** Examine the effects of 1-year burn intervals on bat activity in red pine stands.
- Compare using stationary detectors vs mobile transects to monitor bat activity.

Methods

- Our research took place at Treehaven campus in Northern Wisconsin.
 - Around 2 acre burned units
- Conducted four mobile transects around three units that are burned every year and three units that are not burned at all (control units).
- Set out stationary detectors for six nights in each units.
- We used a Wildlife Acoustics Mini Bat and SM4 Bat for the stationary detectors, and an Echo Meter Touch 2 for the mobile transects.
- Analyzed data using Mann-Whitney Test.



Fig 1. Photos of control units (left) and burn units (right).

Results

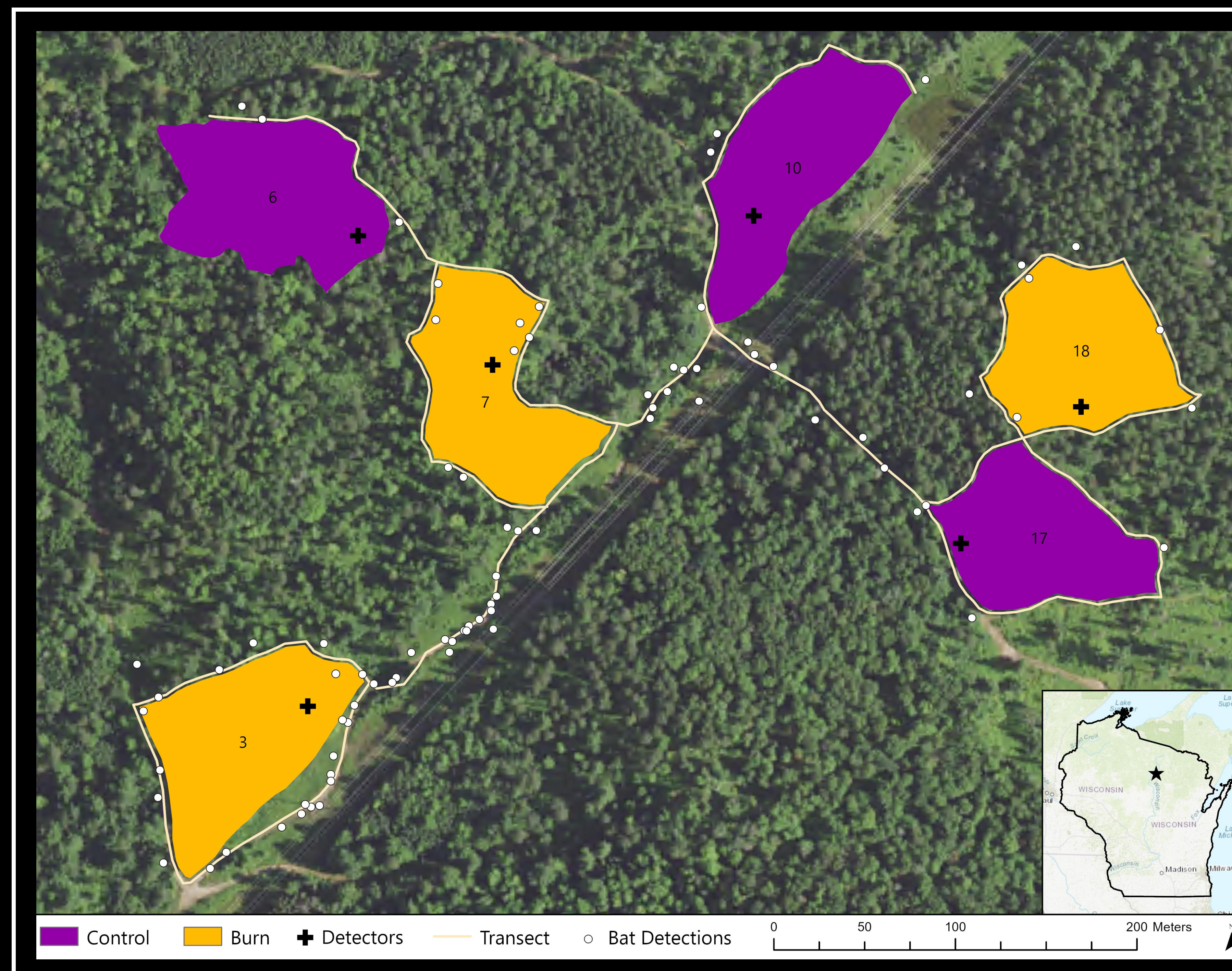


Fig 2. Map of Treehaven burn and control units, the powerline, stationary detector locations, mobile transect route, and mobile bat detections.

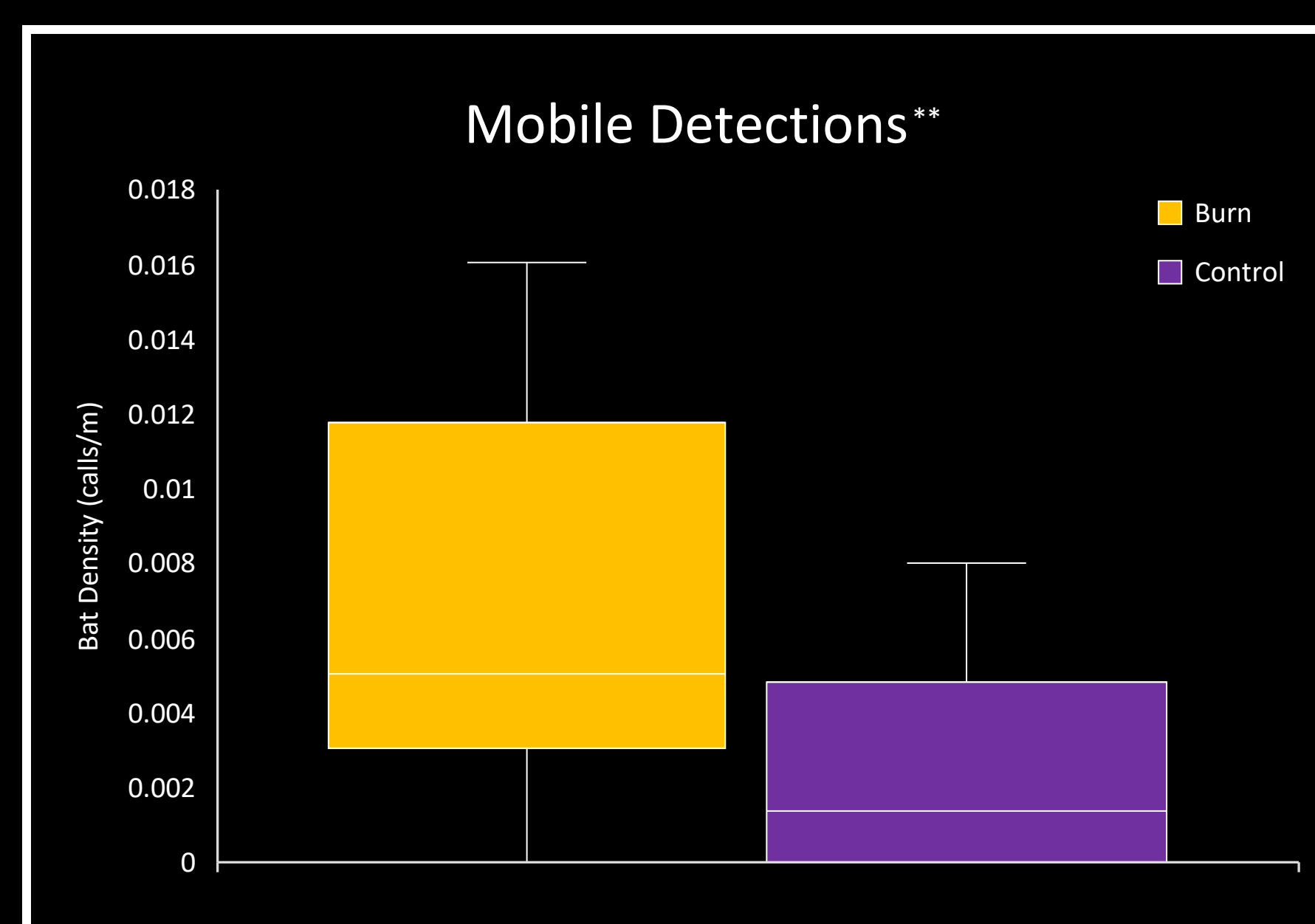


Fig 3. Bat density based on calls from the mobile transects. A Mann-Whitney test yielded a p-value of < 0.01.

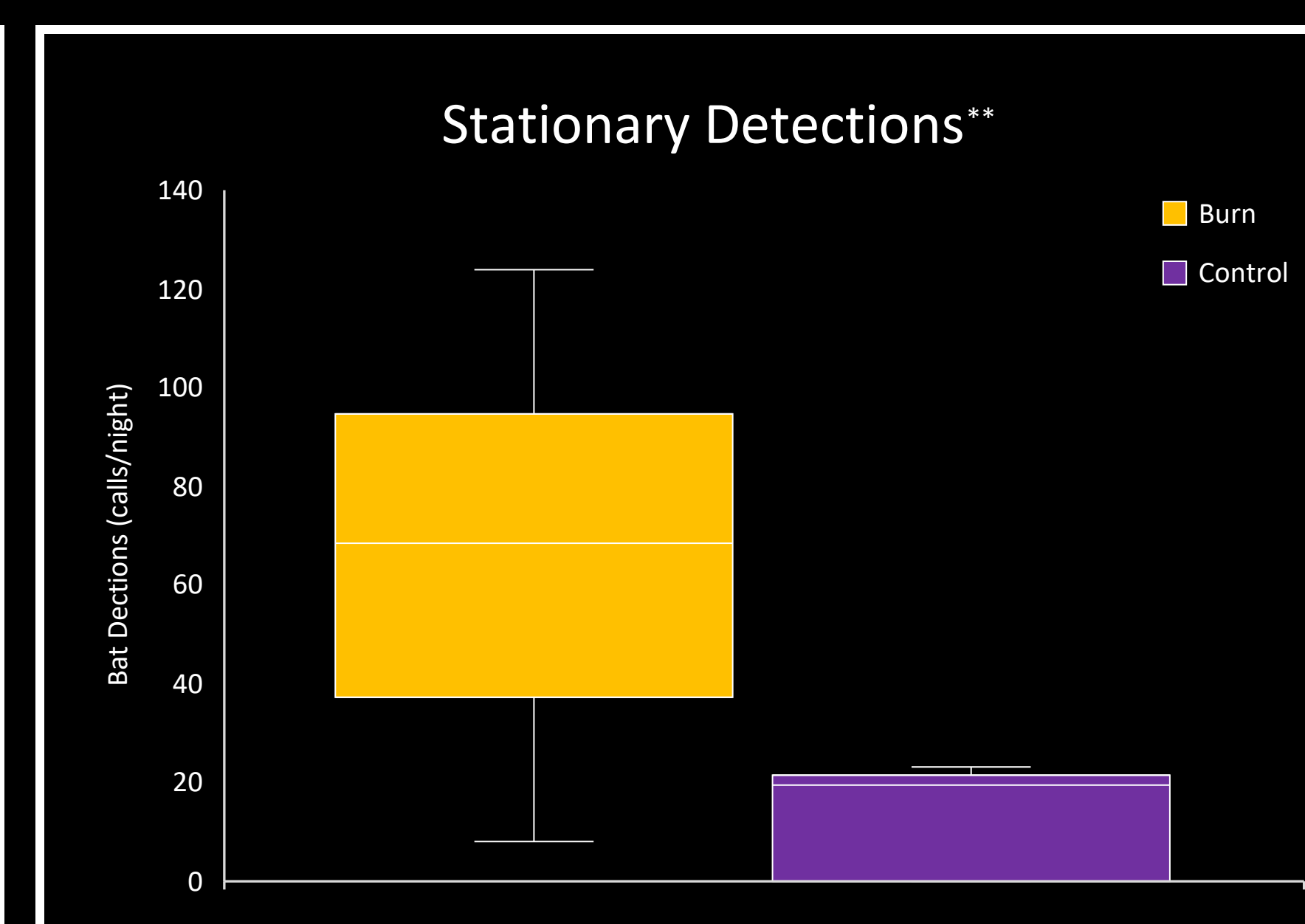


Fig 4. Bat detections from stationary detectors. Using a Mann-Whitney test yielded a p-value of < 0.002.

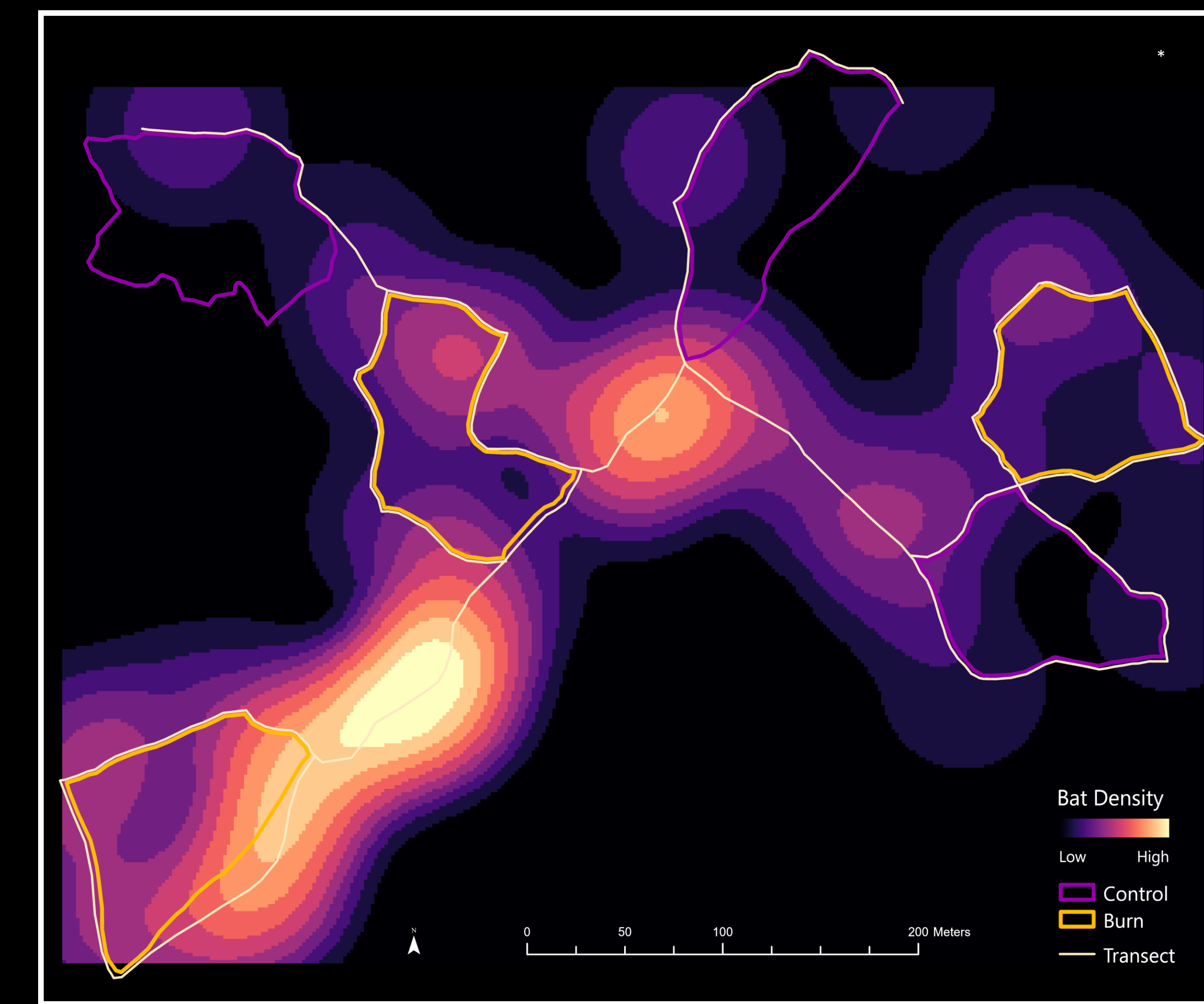


Fig 5. Kernel density estimate along transect. The highest density areas fall on the powerline. Comparing density along the powerline versus in the forested areas with a Mann-Whitney Test yielded a p-value of < 0.02.

Discussion

- Prescribed burning of forests creates good bat habitat.
 - Burning forests with dense mid- and understories can promote bat activity.
 - More research could determine ideal burning interval.
- Stationary detectors collect more total bat calls than mobile transects, but both yielded significant data.
- Powerlines can act as corridors for bats.

Acknowledgements

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